JEL classification: G12.

Measuring Factor Crowding When Leverage Is Observable: Evidence from China's A-Share Market

Jiaheng Wen[a,*]

[a]*Tongji University*

Abstract

Factor crowding is difficult to measure because holdings are rarely observed and return-based proxies may capture several forms of common movement. We investigate whether two proxies identify the same predictive object when leverage is observable. Using six style factors in China's A-share market from 2015–2025, we compare residual-return comovement with a margin-financing-balance proxy. Within-factor comovement between the two measures is limited. Comomentum is not robustly associated with subsequent factor returns; its one-month volatility diagnostic is directionally consistent but the broader second-moment evidence is individually insignificant. The margin-balance proxy is conditionally negatively associated with subsequent factor returns after controlling for factor momentum, balance levels, coverage, and factor fixed effects. In the main six-month specification, a one-standard-deviation increase is associated with approximately a four-percentage-point lower horizon-sum factor return. The association is concentrated in the low-side leg: that leg's own financing intensity predicts its own decay, while the long-side leg contributes little, the effect does not depend on comomentum agreement, and flow-based measures of financing intensity tell the same story. Because rolling formation windows overlap, we use pooled circular-shift permutation inference alongside HAC standard errors. The results support a measurement distinction: return comovement and observable leveraged financing need not be interchangeable measures of factor crowding, and the evidence is associational rather than causal.

Keywords: factor crowding, factor decay, comomentum, margin financing, China, A-share market

1. Introduction

A factor that attracts too much capital should eventually stop working: with limited arbitrage, capital flows into a strategy until the mispricing is gone, and the trades most exposed to a reversal are the crowded ones (Shleifer and Vishny, 1997; De Long et al., 1990; Stein, 2009; Barberis and Shleifer, 2003). What is unsettled is measurement: positions are unobserved, and returns are contaminated by the price pressure crowding itself creates.

*Corresponding author.
Email address: 2790443124@qq.com (Jiaheng Wen)

Lou and Polk (2022) propose a way around this. If many arbitrageurs hold the same stocks, those stocks should comove beyond what common factors explain. Their *comomentum* measure, the average pairwise correlation of three-factor residual returns within the stocks a momentum strategy trades, forecasts lower momentum returns, higher volatility and more negative skewness. It is indirect by construction: it infers positioning from second moments.

We ask whether that inference survives where an alternative is observable. China's A-share market combines a distinct margin-financing institution with tighter short-sale eligibility than the setting in which comomentum was developed. We therefore compare the return- correlation proxy with a balance-based proxy.

The comparison itself is the contribution: a correlation-based and a balance-based proxy can give different predictive answers, complicating any single-proxy interpretation. We construct both for six style factors over 2015–2025 on the full A-share cross-section, including delisted firms. The two measures have limited linear comovement and disagree on predictability: the margin-balance proxy is negatively associated with subsequent factor returns, whereas comomentum is not robustly predictive.

Three concerns shape the design. The first is a mechanical component: a balance-based change measure contains information about the balance level, which correlates with market capitalisation and factor membership. We control for the level throughout; comparing specifications A and C diagnoses, but does not eliminate, this component.

The second concerns factor momentum: factors that performed well may attract financing. We separate the two by orthogonalising the proxy against the factor's past 1-, 6- and 12-month returns, including both simultaneously, and estimating factor-specific regressions. Factor fixed effects identify the pooled coefficient from within-factor variation.

The third concerns inference. With a twelve-month formation window sampled monthly, adjacent observations share eleven twelfths of their data, so we report pooled circular-shift permutation p-values with the same estimator used for the observed statistic.

The paper contributes to two literatures. To the literature on factor crowding and decay (McLean and Pontiff, 2016; Beck et al., 2016; Lou and Polk, 2022), it adds a measurement comparison under a different arbitrage regime. The return-correlation and financing-balance proxies span six factors and are evaluated within the same market. To the literature on China's margin trading (Bian et al., 2026; Xing et al., 2026), it adds a factor-level association between financing balances and subsequent factor returns. The contribution is the measurement comparison and its inferential design, not a claim that either proxy directly observes total investor positions.

2. Data and methodology

2.1. Data

We use the full A-share cross-section from January 2015 to December 2025. Daily prices, trading volumes, turnover, and point-in-time valuation ratios are from Baostock; the universe includes delisted firms. The monthly panel covers 5,247 of the 5,418 stocks in the cleaned universe over 2,674 trading days. Individual-stock margin-financing balances, covering the designated margin-eligible list, are from the Shanghai and Shenzhen exchanges.

We impose a next-trading-day availability rule: the month-end signal uses only balances available by that date; an audit confirms no observation post-dates its signal. The valuation ratios are treated as point-in-time fields based on a 40-stock audit documented in the replication materials.

We exclude stocks that are suspended, have been listed fewer than 180 days, or are flagged ST, and we drop financial firms using the available industry classification (a current-snapshot classification applied to the full sample). Within the monthly cross-section we additionally drop stock-months with more than ten non-trading days, and we apply the same rule when forming the decile legs. The margin-eligible list expands materially over the sample, so we construct factor portfolios on the full cross-section and compute the margin-balance proxy only over constituent stocks, with coverage controls.

Style factors are constructed monthly. Value (VAL) is book-to-market; size (SMB) is negative log float market capitalisation; momentum (MOM) is the compounded return over months $t-12$ to $t-1$; quality (QMJ) is implied return on equity, price-to-book divided by trailing price-to-earnings; low volatility (LOWVOL) is negative realised volatility over the prior 60 trading days; turnover (TURN) is negative mean turnover over the prior 20 days. Decile portfolios are float-capitalisation-weighted, and factor returns are top-decile minus bottom-decile returns, with the long side predicted to earn higher returns.

2.2. Two measures of crowding

Indirect: comomentum.. Following Lou and Polk (2022), for each factor f and month t we take the long and short deciles formed on f's exposure, regress each constituent's daily returns over the trailing 250 trading days on a three-factor benchmark, and compute the average pairwise correlation of residual returns within each leg; the measure is the average across the two legs. Because each leg is capped at 100 randomly selected stocks when necessary, it is a sampled pairwise-correlation measure; the seed is fixed for replication.

Margin-financing-balance proxy.. For each constituent we compute the monthly change in the publicly reported margin-financing– balance scaled by float market capitalisation, $\Delta MB_{i,t}/MV_{i,t}$, and aggregate to the leg level using float-capitalisation weights. The measure is the long-leg aggregate minus the short-leg aggregate. We also retain the long-leg aggregate, the margin level, and coverage. Under exchange disclosure rules, the previous trading day's information is published before the next session, so we use the next trading day as a conservative availability date. Coverage is uneven: the size factor's long leg has mean margin coverage of 0.26 (61 per cent of months below 0.30), against 0.82–0.93 elsewhere. We therefore control for coverage and re-estimate results under coverage floors. This proxy is more directly observed than comomentum, but it is not a pure financing flow, an account-level position, or a causal measure of crowding.

2.3. Empirical design

For each measure $C_{f,t}$ and horizon $h \in \{1, 3, 6, 12\}$ we estimate

$$R^{f}_{t\to t+h} \;=\; \alpha_f \;+\; \beta\, C_{f,t} \;+\; \gamma' X_{f,t} \;+\; \varepsilon_{f,t}, \tag{1}$$

where $R^{f}_{t\to t+h}$ is the horizon sum of the next h monthly long-short returns, starting in $t+1$, α_f is a factor fixed effect, and X collects the factor's own past 1-, 6-, and 12-month returns, the margin level, and coverage. Standard errors are Newey-West with six lags. Because the six factors share a market shock within a month, factor-level clustering is unreliable with so few clusters.

Our central concern is that β reflects factor momentum rather than crowding. We address it three ways: by orthogonalising $C_{f,t}$ against the factor's own past 1-, 6-, and 12-month returns and re-running the regression on the residual; by including crowding and past returns simultaneously; and by estimating the regression separately for each factor.

Table 1: Comomentum and margin crowding: correlation by factor

Factor	N	Pearson	Spearman	Mean comomentum	Mean margin-balance proxy
VAL	120	-0.269	-0.159	0.02307	-0.0008
SMB	120	0.266	0.249	0.00638	-0.0001
MOM	120	-0.168	-0.151	0.02656	0.0010
QMJ	120	0.052	-0.013	0.01439	0.0002
LOWVOL	120	0.008	0.009	0.02970	-0.0019
TURN	120	-0.252	-0.192	0.02043	-0.0037
Pooled	720	-0.141	-0.144	0.02009	-0.0009
Pooled (factor-demeaned)	720	-0.093	-0.073	0.00000	0.0000

For each permutation we shift the measure within factor and re-estimate the same pooled specification, preserving factor fixed effects, controls, the sample, and the HAC procedure. We use the finite-permutation correction $(1 + B_{\geq})/(1 + B)$ for the reported p-values and treat them as the primary inferential quantities; HAC p-values are shown for comparison. We validate the correction on synthetic panels with no crowding effect by construction, and since a null result is only informative if the design has power, we compute minimum detectable effects by block bootstrap; both checks are in the replication materials.

The primary evidence family is the two proxies crossed with Layers 1 and 3 and horizons $h \in \{1, 3, 6\}$; horizon 12 and the remaining analyses are auxiliary.

3. Results

3.1. The two measures of crowding are only weakly related

The two measures have limited linear comovement. Table 1 reports the within-factor correlation between comomentum and the margin-balance proxy. It ranges from -0.25 for turnover (and -0.27 for value) to $+0.27$ for size, and is near zero for low-volatility. Pooling the six factors gives -0.14, or -0.09 after removing factor-specific means. These correlations may also reflect timing, coverage, nonlinear dependence, or sampling noise.

3.2. Margin-balance proxy predicts decay; comomentum does not

Table 2 reports the primary coefficients and permutation p-values; HAC t-statistics are retained in the replication outputs. Comomentum has no robust predictive association with subsequent factor returns in this sample and design: permutation p-values run from 0.36 to 0.95, and the point estimates change sign across horizons. The margin-balance proxy, by contrast, is negatively associated with subsequent returns.

The pattern across specifications is informative. With no controls (column A) the margin-balance coefficient is negative but imprecise. Adding the factor's own past 1-, 6- and 12-month returns (column B) increases its magnitude; at $h = 6$, β is about -0.034 (permutation $p = 0.010$). Controlling for the margin level (C) and coverage (D) leaves the estimates essentially unchanged, diagnosing but not eliminating concerns about the level component of balance changes.

The magnitude is economically material: in specification D at $h = 6$, a one-standard-deviation increase in the margin-balance proxy accompanies approximately a 4.0 percentage point lower horizon-sum long-short return over six months (3.3 points per within-factor standard deviation). This is a conditional predictive association, not an estimate of a causal effect.

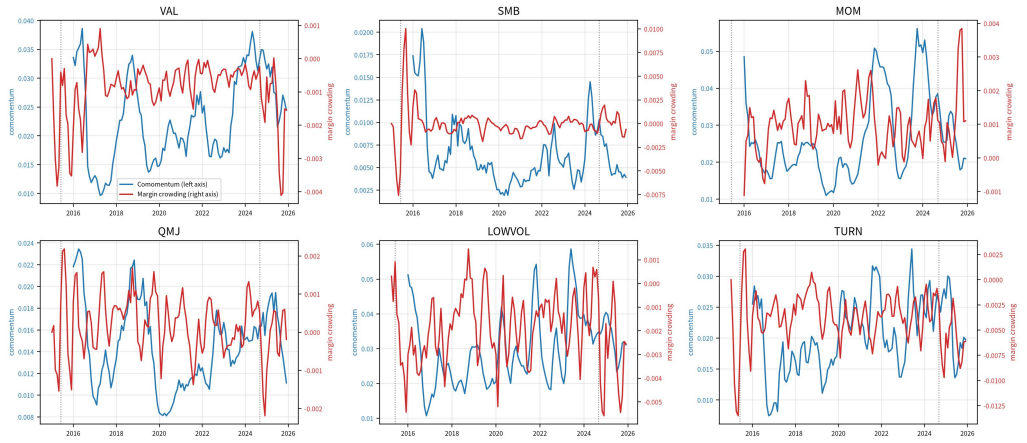


Figure 1: The two crowding measures by factor, monthly, three-month rolling mean. Comomentum (blue, left axis) is the average pairwise correlation of three-factor residual returns within the factor's long and short legs. Margin crowding (red, right axis) is the float-weighted change in margin financing balances on the long leg less the short leg, scaled by float market capitalisation. Dotted vertical lines mark June 2015 and September 2024. The two series are plotted on separate axes because their units differ by one to two orders of magnitude.

3.3. The signal is in the low-side leg

China's margin system finances long positions almost exclusively, so we ask which leg carries the association. In Table 3, Panel A regresses each leg's horizon-sum return on the total margin-balance measure with specification-D controls and sample, so the long-short coefficient equals the long-leg coefficient minus the short-leg coefficient. At $h = 6$ the long leg contributes nothing ($\beta = 0.0005$, permutation $p = 0.97$) while the short leg carries the entire effect ($\beta = 0.0406$, $p = 0.010$). Panel B pairs each leg with its own financing measure: the low-side leg's financing intensity predicts its own decay strongly ($\beta = -0.0517$ at $h = 6$, $p = 0.010$); the long-side leg shows the same sign but is insignificant ($\beta = -0.0162$, $p = 0.14$). The association reflects low-side financing intensity rather than a fragile spread artifact. Comomentum shows no leg-level pattern.

3.4. Flows versus stocks

The balance-change measure is a stock concept: it captures where financing stands. Daily financing purchases and repayments are also disclosed, so we construct a flow-based variant: the monthly sum of financing purchases scaled by float market capitalisation, aggregated as before, with its trailing twelve-month mean as a level control. Net flows, purchases less repayments, are available for Shanghai only. Table 4 shows that both flow measures are negatively associated with subsequent factor returns; the Shanghai net-flow measure is significant at all four horizons (permutation $p = 0.016$–0.019). When flow and balance-change measures compete in the same specification, the balance-change measure remains significant at every horizon, while purchase flows retain independent content at one month ($p = 0.016$) and net flows at six months ($p = 0.034$). The two margin-based measures are only moderately correlated (0.44–0.50), so stocks and flows capture partly distinct signals.

Table 2: Crowding and future factor returns

Panel A: coefficient (stars from permutation p)					
Measure	h	A	B	C	D
Comomentum	1	0.0019	0.0021	0.0031	0.0036
Comomentum	3	0.0011	0.0037	0.0068	0.0082
Comomentum	6	0.0014	0.0059	0.0113	0.0131
Comomentum	12	-0.0211	-0.0183	-0.0107	-0.0102
Margin-balance proxy	1	-0.0087***	-0.0137***	-0.0150***	-0.0157***
Margin-balance proxy	3	-0.0115	-0.0226***	-0.0264***	-0.0284***
Margin-balance proxy	6	-0.0177	-0.0338***	-0.0383***	-0.0400***
Margin-balance proxy	12	-0.0190	-0.0377*	-0.0469**	-0.0479**

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel B: finite permutation p-value					
Measure	h	A	B	C	D
Comomentum	1	0.571	0.607	0.355	0.299
Comomentum	3	0.949	0.724	0.590	0.486
Comomentum	6	0.930	0.853	0.539	0.480
Comomentum	12	0.639	0.667	0.781	0.792
Margin-balance proxy	1	0.008	0.009	0.009	0.009
Margin-balance proxy	3	0.376	0.010	0.010	0.010
Margin-balance proxy	6	0.202	0.010	0.010	0.010
Margin-balance proxy	12	0.602	0.052	0.010	0.010

3.5. The return-correlation proxy is not merely a failed implementation

The obvious objection is that our implementation of comomentum fails, rather than the proxy itself. Lou and Polk (2022) also document that comomentum forecasts higher realised volatility and more negative skewness. Table 5 shows some directional consistency in China: the one-month volatility coefficient has permutation $p = 0.059$, while the skewness results are not individually significant. This is supportive diagnostic evidence, not a full replication.

The margin-balance proxy is negatively associated with returns and has no reliable relation to the second moments. The two proxies therefore need not be interchangeable: comomentum tracks residual-return comovement, whereas the margin-balance proxy tracks a balance-based financing signal.

3.6. Separating crowding from factor momentum

Table 6 reports the defenses against the interpretation that β is factor momentum. Orthogonalising the proxy against past returns and the margin level leaves the primary margin-balance association intact, and including the proxy and past returns simultaneously gives the same qualitative result. Factor-specific slopes are reported only as directional evidence; with six factors, they do not provide a reliable stand-alone test.

3.7. The effect is in the centre, not the tail

A natural competitor to the mean-shift story is that the margin-balance proxy raises crash risk; the auxiliary evidence does not establish that pattern. Table 8 reports pooled quantile regressions

Table 3: Leg decomposition of the crowding association

Panel	Measure	Return leg	h	β	HAC t	Perm. p	N
A	Comomentum	Long-short	1	0.0036	0.99	0.299	699
	Comomentum	Long leg	1	-0.0017	-0.60	0.542	699
	Comomentum	Short leg	1	-0.0052	-1.66	0.103	699
A	Comomentum	Long-short	6	0.0131	0.87	0.480	669
	Comomentum	Long leg	6	0.0049	0.45	0.775	669
	Comomentum	Short leg	6	-0.0082	-0.59	0.755	669
A	Margin-balance proxy	Long-short	1	-0.0157***	-4.30	0.009	699
	Margin-balance proxy	Long leg	1	-0.0015	-0.64	0.551	699
	Margin-balance proxy	Short leg	1	0.0142***	3.79	0.009	699
A	Margin-balance proxy	Long-short	6	-0.0400***	-3.91	0.010	669
	Margin-balance proxy	Long leg	6	0.0005	0.07	0.971	669
	Margin-balance proxy	Short leg	6	0.0406***	4.02	0.010	669
B	Margin-balance proxy, long leg	Long leg	1	-0.0085*	-2.25	0.065	699
B	Margin-balance proxy, short leg	Short leg	1	-0.0229***	-4.51	0.009	699
B	Margin-balance proxy, long leg	Long leg	6	-0.0162	-1.89	0.137	669
B	Margin-balance proxy, short leg	Short leg	6	-0.0517***	-4.74	0.010	669
C	Margin-balance proxy, long leg	Long → short leg	1	-0.0088	-2.21	0.131	699
C	Margin-balance proxy, short leg	Short → long leg	1	-0.0012	-0.51	0.710	699
C	Margin-balance proxy, long leg	Long → short leg	6	-0.0254*	-2.65	0.078	669
C	Margin-balance proxy, short leg	Short → long leg	6	-0.0069	-1.01	0.490	669

Notes: Panel A regresses each leg's horizon-sum return on the total margin-balance or comomentum measure, with the identical controls and sample as specification D, so the long-short coefficient equals the long-leg coefficient minus the short-leg coefficient. Panel B pairs each leg's return with that leg's own financing measure; Panel C pairs legs across sides. Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

with factor fixed effects, and Table 7 compares the two crowding groups' distributions. The largest negative coefficients occur near the median and upper quantiles, while the left-tail coefficient is not significant. These results do not constitute a formal test of quantile-coefficient equality or a point-in-time crash threshold.

3.8. Robustness and scope of the null result

Table 9 splits the sample, changes the weighting, and varies the construction window. The margin-balance association remains detectable at the 0.10 level in 2015–2018 ($p = 0.194$) but not in 2019–2021 ($p = 0.111$) or 2022–2025 ($p = 0.149$), although all three subsample estimates are negative. It survives equal weighting ($p = 0.008$) and weakens when the balance change is measured over six or twenty-four months. Restricting the measure to either leg alone removes the one-month association ($p = 0.67, 0.22$), consistent with the leg decomposition. These results show directional and design sensitivity; they should not be described as universal robustness.

Further checks support the main specification. Compounding returns leaves the coefficient slightly larger (-0.042 at $h = 6$, $p = 0.010$). Excluding the size factor, whose margin coverage is thinnest, strengthens the association (-0.049, $p = 0.010$), and coverage floors of 0.30 or 0.50 leave it intact (-0.046 at $h = 6$; $p = 0.022, 0.025$). The margin signal does not depend on comomentum agreement: coefficients in months above and below the factor-specific comomentum median are similar (-0.036 and -0.035 at $h = 6$), and the interaction is insignificant ($p = 0.20$). The two proxies act as independent signals rather than complementary ones.

Because one result is a null, its informativeness matters: minimum detectable effects from the block bootstrap are about 11–14 per cent annualised for comomentum. The null is informative

Table 4: Financing flows versus balance changes

Measure	h	β	HAC t	Perm. p	N
Financing purchases (buy flow)	1	-0.0330**	-3.72	0.016	644
Financing purchases (buy flow)	3	-0.0625**	-3.82	0.016	632
Financing purchases (buy flow)	6	-0.0978*	-3.95	0.051	614
Financing purchases (buy flow)	12	-0.0521	-1.48	0.623	578
Net financing flows (SSE)	1	-0.0171**	-4.09	0.016	644
Net financing flows (SSE)	3	-0.0296**	-3.79	0.016	632
Net financing flows (SSE)	6	-0.0605**	-5.74	0.017	614
Net financing flows (SSE)	12	-0.0517**	-2.70	0.019	578

Notes: Both measures are leg-level float-weighted monthly financing intensities (long leg minus short leg) with specification-D controls and the measure's own trailing twelve-month mean as an additional level control; net flows are available for Shanghai only. Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

relative to effects above this threshold.

4. Conclusion

Two proxies for factor crowding, applied to the same factors in the same market, give substantially different predictive results. The return-correlation proxy has no robust association with subsequent factor returns in this sample and design. The margin-balance proxy is negatively associated with subsequent returns after controls for factor momentum, the balance level, coverage, factor fixed effects, and pooled circular-shift permutation inference. The association is concentrated in the low-side leg, whose own financing intensity predicts its own decay, it does not depend on comomentum agreement, and flow-based financing measures echo the balance-based result. The result is conditional and associational, not a causal estimate or an account-level holdings result.

The evidence supports a measurement distinction. Comomentum provides a residual-return comovement signal with limited directional evidence in the second-moment diagnostics, while the margin-balance proxy provides the stronger return-predictive association. The two proxies should therefore be reported side by side rather than treated as interchangeable. The subsample and window results show that the association is not universal, and the tail analysis does not establish disproportionate crash risk.

Trailing-window measures sampled monthly create overlapping observations; the pooled circular-shift procedure preserves each factor's measure persistence and is the primary inferential procedure here. Future work should use historical industry classifications, account- or position-level data, and quasi-experimental variation in margin eligibility or disclosure.

Table 5: Crowding and the second moments of factor returns

Panel A: coefficient			
Measure	h	Volatility	Skewness
Comomentum	1	0.0014*	-0.0909
Comomentum	3	0.0009	-0.1312
Comomentum	6	0.0006	-0.1242
Margin-balance proxy	1	-0.0002	-0.0236
Margin-balance proxy	3	-0.0006*	-0.0390
Margin-balance proxy	6	-0.0005	-0.0326
Panel B: permutation p-value			
Measure	h	Volatility	Skewness
Comomentum	1	0.059	0.288
Comomentum	3	0.216	0.310
Comomentum	6	0.407	0.345
Margin-balance proxy	1	0.737	0.602
Margin-balance proxy	3	0.095	0.302
Margin-balance proxy	6	0.115	0.239

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Separating crowding from factor momentum

Measure	Layer	h	β	HAC t	Perm. p
Comomentum	Layer 1: orthogonalised residual	1	0.0031	0.87	0.374
Comomentum	Layer 1: orthogonalised residual	3	0.0068	0.72	0.610
Comomentum	Layer 1: orthogonalised residual	6	0.0117	0.72	0.706
Comomentum	Layer 1: orthogonalised residual	12	-0.0077	-0.27	0.906
Comomentum	Layer 3: all controls simultaneously	1	0.0031	0.87	0.355
Comomentum	Layer 3: all controls simultaneously	3	0.0068	0.76	0.590
Comomentum	Layer 3: all controls simultaneously	6	0.0113	0.75	0.539
Comomentum	Layer 3: all controls simultaneously	12	-0.0107	-0.43	0.781
Margin-balance proxy	Layer 1: orthogonalised residual	1	-0.0150***	-3.95	0.009
Margin-balance proxy	Layer 1: orthogonalised residual	3	-0.0260***	-3.69	0.010
Margin-balance proxy	Layer 1: orthogonalised residual	6	-0.0392***	-3.76	0.010
Margin-balance proxy	Layer 1: orthogonalised residual	12	-0.0469*	-2.69	0.052
Margin-balance proxy	Layer 3: all controls simultaneously	1	-0.0150***	-4.09	0.009
Margin-balance proxy	Layer 3: all controls simultaneously	3	-0.0264***	-4.10	0.010
Margin-balance proxy	Layer 3: all controls simultaneously	6	-0.0383***	-3.78	0.010
Margin-balance proxy	Layer 3: all controls simultaneously	12	-0.0469**	-2.98	0.010
Comomentum	Layer 2: within-factor time series	1	0.0018	0.32	0.776
Comomentum	Layer 2: within-factor time series	1	-0.0062	-0.69	0.551
Comomentum	Layer 2: within-factor time series	1	-0.0215	-1.98	0.118
Comomentum	Layer 2: within-factor time series	1	0.0063	0.50	0.723
Comomentum	Layer 2: within-factor time series	1	0.0108	1.10	0.345
Comomentum	Layer 2: within-factor time series	1	0.0142	2.08	0.109
Comomentum	Layer 2: cross-factor mean slope	1	0.0009	0.17	—
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0130	-1.41	0.181
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0406**	-3.78	0.019
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0042	-0.47	0.672
Margin-balance proxy	Layer 2: within-factor time series	1	0.0026	0.25	0.874
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0193**	-2.98	0.042
Margin-balance proxy	Layer 2: within-factor time series	1	-0.0186	-1.05	0.328
Margin-balance proxy	Layer 2: cross-factor mean slope	1	-0.0155	-2.54	—

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Crowding and the distribution of factor returns

Measure	Factor	Crowding	Mean	SD	Skew.	Kurt.	Crash freq.
Comomentum	VAL	High	0.0127	0.0582	0.38	0.83	0.02
	VAL	Low	-0.0089	0.0748	-0.40	0.16	0.07
	SMB	High	0.0091	0.0635	0.13	0.04	0.03
	SMB	Low	-0.0021	0.0651	-0.36	0.90	0.08
	MOM	High	0.0007	0.0684	-0.43	0.42	0.07
	MOM	Low	0.0143	0.0734	-0.08	0.91	0.03
	QMJ	High	-0.0057	0.0514	-1.23	2.68	0.07
	QMJ	Low	0.0134	0.0545	-0.37	0.67	0.03
	LOWVOL	High	0.0087	0.0907	-0.07	1.00	0.07
	LOWVOL	Low	0.0113	0.0685	-0.29	0.16	0.03
	TURN	High	0.0172	0.0960	-0.83	2.59	0.05
	TURN	Low	0.0225	0.0660	0.26	0.81	0.02
Margin-balance proxy	VAL	High	0.0002	0.0696	-0.13	0.67	0.03
	VAL	Low	0.0013	0.0832	-0.54	2.51	0.08
	SMB	High	0.0071	0.0663	0.25	0.32	0.05
	SMB	Low	0.0136	0.0895	1.59	6.44	0.06
	MOM	High	-0.0028	0.0684	-0.30	0.40	0.07
	MOM	Low	0.0179	0.0725	-0.20	1.06	0.03
	QMJ	High	0.0023	0.0534	-0.30	-0.03	0.05
	QMJ	Low	0.0029	0.0544	-0.90	2.71	0.06
	LOWVOL	High	0.0112	0.0835	-0.06	1.35	0.06
	LOWVOL	Low	0.0094	0.0825	-0.26	0.59	0.05
	TURN	High	0.0160	0.0715	-0.58	2.05	0.03
	TURN	Low	0.0191	0.0993	-0.65	1.60	0.08

Table 8: Quantile regressions: does crowding move the tail more than the centre?

Measure	$q = 0.10$	$q = 0.25$	$q = 0.50$	$q = 0.75$	$q = 0.90$
Comomentum	0.0010	-0.0006	-0.0037	0.0037	0.0098
Margin-balance proxy	0.0011	-0.0020	-0.0093**	-0.0097*	-0.0107
Permutation p-values					
Comomentum	0.798	0.866	0.361	0.429	0.151
Margin-balance proxy	0.916	0.710	0.031	0.061	0.145

Table 9: Robustness

Check	Grouping	Measure	β	HAC t	Perm. p	N
Subsample	2015–2018 leverage cycle	Comomentum	-0.0026	-0.51	0.583	216
Subsample	2015–2018 leverage cycle	Margin-balance proxy	-0.0046	-1.25	0.194	273
Subsample	2019–2021 foreign inflows	Comomentum	0.0043	0.46	0.750	216
Subsample	2019–2021 foreign inflows	Margin-balance proxy	-0.0241	-3.04	0.111	216
Subsample	2022–2025 retail margin	Comomentum	0.0101	1.72	0.128	282
Subsample	2022–2025 retail margin	Margin-balance proxy	-0.0103	-2.32	0.149	282
Equal-weighted	Equal-weighted	Comomentum	0.0000	0.01	0.992	714
Equal-weighted	Equal-weighted	Margin-balance proxy	-0.0072***	-3.01	0.008	771
Window	Balance change, 6 months	Margin-balance proxy	-0.0076	-1.33	0.185	771
Window	Balance change, 24 months	Margin-balance proxy	-0.0053	-1.50	0.244	771
Variant	Long leg only	Margin-balance proxy, long leg	-0.0011	-0.61	0.672	771
Variant	Short leg only	Margin-balance proxy, short leg	0.0027	1.63	0.218	771

Notes: Stars use finite circular-shift permutation p-values; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Author declarations

Data availability

The daily market and valuation data are obtained from Baostock, and the margin-financing data are obtained from public exchange disclosures. Redistribution of raw provider data may be subject to upstream terms. The analysis code, configuration, evidence manifest, run manifest, derived-data documentation, and derived summary outputs are publicly available at `https://github.com/WEN365-1/a-share-factor-crowding-research`; raw provider data are documented in a companion repository (`https://github.com/WEN365-1/a-share-factor-crowding-raw-data`), subject to the providers' terms.

Funding

No external funding is reported.

Declaration of competing interests

The author declares no competing interests.

CRediT authorship contribution statement

Jiaheng Wen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Declaration of generative AI use

During the preparation of this work, the author used DeepSeek, a large language model, for research assistance and language polishing. The author reviewed and edited the content, verified the reported results, and takes full responsibility for the content of the published article.

Acknowledgements

The author is responsible for all remaining errors.

References

Barberis, N., Shleifer, A., 2003. Style investing. Journal of Financial Economics 68, 161–199. doi:10.1016/S0304-405X(03)00064-3.

Beck, N., Hsu, J., Kalesnik, V., Kostka, H., 2016. Will your factor deliver? an examination of factor robustness and implementation costs. Financial Analysts Journal 72, 63–74. doi:10.2469/faj.v72.n5.6.

Bian, J., Da, Z., He, Z., Lou, D., Shue, K., Zhou, H., 2026. The drivers and implications of retail margin trading. The Journal of Finance 81, 2217–2270. doi:10.1111/jofi.70049.

De Long, J.B., Shleifer, A., Summers, L.H., Waldmann, R.J., 1990. Noise trader risk in financial markets. Journal of Political Economy 98, 703–738. doi:10.1086/261898.

Lou, D., Polk, C., 2022. Comomentum: Inferring arbitrage activity from return correlations. The Review of Financial Studies 35, 3272–3302. doi:10.1093/rfs/hhab117.

McLean, R.D., Pontiff, J., 2016. Does academic research destroy stock return predictability? The Journal of Finance 71, 5–32. doi:10.1111/jofi.12365.

Shleifer, A., Vishny, R.W., 1997. The limits of arbitrage. The Journal of Finance 52, 35–55. doi:10.1111/0022-1082.00001.

Stein, J.C., 2009. Presidential address: Sophisticated investors and market efficiency. The Journal of Finance 64, 1517–1548. doi:10.1111/j.1540-6261.2009.01472.x.

Xing, S., Luo, Y., Li, T., 2026. PMO factor crowding and returns: Evidence from china. Applied Economics Letters 33, 987–993. doi:10.1080/13504851.2024.2401517.